BOKPOINTS: Functional Requirements Document (FRD)

Version 0.2 (draft), based on PRD v0.1 and the product owner's answers to the open decisions. Product owner and product manager: A. Prepared with Claude. Status: planning, no build started.

1. Purpose and conventions

This document says exactly what BOKPOINTS must do. The PRD explains why and for whom; this FRD explains what the system does, screen by screen and rule by rule. It is written so that one builder, or an AI coding agent working from AGENT.md, can implement from it.

Detail level: the MVP is specified in full. v1.1 and v2 are outlined only and will be detailed before their phase begins.

Requirement IDs: `FR-<MODULE>-<number>`, for example `FR-PTS-3`.

Priority words:

- **MUST:** required for the phase it is tagged with.
- **SHOULD:** strongly wanted; cut only if time forces it.
- **COULD:** nice to have.

Phase tags: [MVP], [v1.1], [v2]. A requirement with no tag is [MVP].

Modules: AUTH (accounts and sign-in), CUS (customer dashboard), STO (store dashboard), ADM (admin panel), PTS (points engine), CRD (credentials), BIL (billing), NTF (notifications), API (API and integrations), AUD (audit and fraud), PRV (privacy and consent), LOC (localization), NFR (non-functional).

2. Roles and permissions

Capability	Admin	Store owner	Store manager	Cashier	Customer
Create, suspend and offboard stores	Yes	No	No	No	No

Capability	Admin	Store owner	Store manager	Cashier	Customer
Manage store profile and billing	Yes (support)	Yes	No	No	No
Add and remove staff	Yes (support)	Yes	Yes (cashiers only)	No	No
Set points rules	Yes (support)	Yes	Yes	No	No
Scan customer and award points	No	Yes	Yes	Yes	No
Manual point adjustment (never below zero)	Yes (support)	Yes	Yes	No	No
Fraud clawback (may go below zero)	Yes	No	No	No	No
Manage cashier devices	Yes (support)	Yes	Yes	No	No
View store customer list	No (support lookup only)	Yes	Yes	Limited (name and balance at scan)	No
Manage products and discounts	No	Yes	Yes	No	No
Send alerts and campaigns [v1.1]	No	Yes	Yes	No	No
View own points and history	No	No	No	No	Yes
Manage own consent and account	No	Own	Own	Own	Yes
View audit logs	Yes	Store-level	Store-level	No	Own history only

A person can hold different roles in different stores. Admin accounts are separate from store and customer accounts.

3. Core concepts and data model

Entity	Purpose	Key fields
Customer	One global person account	id, name, email, phone (optional, not verified in the MVP), verification status, language, created date, status
Credential	A way to identify a customer at the till	id, customer id, type (app QR, NFC card, wallet pass), identifier, device public key, valid until, status (active, lost, revoked), created date
Store	A participating business	id, name, business ID, address, contact, VAT details, status, plan, created date
Store member	A staff account at a store	user id, store id, role (owner, manager, cashier), status
Cashier device	A till device registered to a store	id, store id, name, status (active, revoked), last sync, registered by
Customer-store link	Connects a customer to a store	customer id, store id, joined date, marketing consent for this store, status (active, under fraud review)
Points rule	The store's earning rules	store id, rule version, points per euro, minimum purchase, maximum points per transaction, rounding, active from date
Points ledger entry	One change to a balance (append-only)	id, customer id, store id, type (earn, redeem, adjust, expire, fraud clawback), points (positive or negative), remaining points and expiry date (earn entries), amount in cents, rule version, reason, staff id, device id, scan time, idempotency key, created date

Entity	Purpose	Key fields
Points balance	Current total per customer and store	derived from the ledger; cached for speed
Rejected offline award	An offline award the server did not accept at sync	id, store id, device id, customer id, amount, scan time, reason, reviewed by
Product [v1.1]	Item in the store shop	id, store id, name, description, price, image, stock note, active
Discount [v1.1]	A price reduction on a product	product id, type, value, start and end date, status
Reward [v1.1]	Something a customer can redeem with points	store id, name, point cost, active, limits
Campaign / alert [v1.1]	A message from a store to customers	store id, audience, message, schedule, status, results
Subscription	A store's paid plan	store id, plan, status, billing dates, first failed payment date, provider references
Audit log entry	Record of a sensitive action	who, what, when, which store, before and after values

Rules that hold across the model:

- The ledger is append-only. Entries are never edited or deleted. There are no reversals: a manual adjustment (see 4.5) corrects a mistake, and only an admin can apply a fraud clawback (see 4.6).
- Money is stored as whole cents in euros. Points are whole numbers.
- A customer's balance at a store is the sum of that customer's ledger entries for that store, expiry entries included.
- Points belong to the store that issued them and cannot be used elsewhere.
- Points expire 3 months after the day they were earned. The oldest points are used first.
- A balance never goes below zero, except through an admin fraud clawback.

4. Key flows (MVP)

4.1 Admin onboards a store

1. Admin creates the store record: name, business ID, address, contact person, email, VAT details, plan.
2. System sends an invitation email to the owner with a secure link.
3. Owner accepts, sets up sign-in, and lands in the store dashboard in setup mode.
4. Owner completes the store profile, sets the points rule, and adds staff.
5. Subscription starts according to the plan (trial or immediate billing).
6. Store status becomes Active once setup and billing are complete.

Store status lifecycle: Invited, Setup, Active, Paused (from day 14 of a failed payment), Suspended (from day 30), Offboarded. A reminder goes to the owner on day 7 without changing the status (see FR-BIL-4).

4.2 Customer signs up

1. Customer installs the app from the Apple App Store or Google Play, or opens the web app, and enters an email address.
2. Customer verifies with a one-time code or link sent by email.
3. Customer accepts the terms and privacy notice, and chooses a language. A phone number can be added but is not verified in the MVP.
4. Customer lands on the dashboard with a personal QR code and an empty store list.

4.3 Cashier awards points (core flow)

1. Cashier opens the Cashier screen (an installable web app) and signs in once per shift or device.
2. Customer shows the QR code in the app (later: taps an NFC card). The QR code works with or without a connection.
3. Cashier scans it with the device camera.
4. The device checks the code. Online, the system resolves the credential to the customer and shows the first name and the current balance at this store. Offline, the device verifies the code on its own and shows the first name only; no balance is shown.
5. Cashier enters the purchase amount in euros.

6. The screen shows the amount and the points it will award, and the cashier confirms. There is no undo after this step.
7. Online: the system writes a ledger entry and updates the balance, and both screens show the confirmation. Offline: the device saves the award in a local queue, shows "Saved offline", and sends it when the connection returns; the customer's balance updates after that sync.

If this is the customer's first purchase at this store, the system also creates the customer-store link when the award is recorded. The customer sees a first-time notice the next time they open the app, with the store's marketing consent shown as off until the customer opts in.

4.4 Cashier registers a customer without the app [MVP]

1. Cashier enters the customer's email address.
2. System sends the customer an invitation to create an account and claim any points. SMS is not used in the MVP.
3. Points earned before sign-up are held on a pending account and become visible once the customer verifies the account.
4. Pending points expire after 3 months like any other points. If the customer never claims the account, the pending record, including the email address, is deleted after 1 year.

4.5 Manual adjustment

1. Owner or manager opens a customer in the store customer list.
2. They choose Add or Remove points, enter the amount and a required reason. Removing points cannot take the balance below zero.
3. System writes an adjustment ledger entry and an audit log entry.

4.6 Fraud clawback (admin only)

1. After a fraud case is confirmed, the admin opens the customer-store record.
2. Admin chooses Fraud clawback, enters the points to remove and a case note. This is the only action that can take a balance below zero.
3. System writes a clawback ledger entry, marks the customer-store link as under fraud review, and writes an audit log entry. Later earnings first offset a negative balance (proposed).

4.7 Redemption [v1.1]

1. Customer chooses a reward in the app or asks at the till.
2. Cashier scans the customer, selects the reward, and confirms.
3. System checks the balance, writes a redeem entry using the oldest points first, and shows the new balance.

4.8 Offline awards and sync [MVP]

1. The Cashier screen keeps working without a connection. It holds the store's points rule, the list of revoked credentials, and the keys it needs to check QR codes.
2. Each offline award is saved in an encrypted local queue with its scan time, amount, points, rule version, device ID and idempotency key.
3. When the connection returns, the device sends the queue automatically and retries until every item is confirmed.
4. The server accepts an award if, at the scan time, the credential was valid, the store was not suspended, the QR proof had not been used before, and the amount was within the offline limits.
5. An accepted award is written to the ledger with its original scan time, and the customer's balance updates.
6. A rejected award is not written to the ledger. It appears in the store's rejected-awards list with the reason, and the owner can add points manually if the purchase was genuine.

5. Functional requirements

5.1 AUTH: accounts and sign-in

- **FR-AUTH-1 (MUST):** Customers sign up and sign in with an email address verified by a one-time code or link. A phone number is optional and is not used for verification in the MVP.
- **FR-AUTH-2 (MUST):** Store staff and admins sign in with email and a secure credential. Multi-factor authentication is required for admin and store owner accounts.
- **FR-AUTH-3 (MUST):** Email is unique per customer. A duplicate sign-up is blocked, or merged only after verification.

- **FR-AUTH-4 (MUST):** Sessions expire after a configurable idle time. Cashier devices keep a longer session so they can work offline, and the owner can revoke any device; a revoked device stops working at its next sync.
- **FR-AUTH-5 (MUST):** Users can sign out everywhere and reset access securely.
- **FR-AUTH-6 (MUST):** Access to every screen and action is enforced on the server by role and by store; hiding buttons is never the only control.

5.2 CUS: customer dashboard

- **FR-CUS-1 (MUST):** Show the customer's personal QR code on the home screen, large and readable, with screen brightness guidance. The code works without a connection.
- **FR-CUS-2 (MUST):** List every store where the customer has points, with store name, current balance, and last activity.
- **FR-CUS-3 (MUST):** Show per-store history: date, type (earned, redeemed, adjusted, expired), points, and resulting balance.
- **FR-CUS-4 (MUST):** Show a confirmation after each award. An award saved offline at the till shows as pending until that till syncs.
- **FR-CUS-5 (MUST):** Let the customer manage profile data, language, and notification and marketing consent per store.
- **FR-CUS-6 (MUST):** Let the customer export their data and delete their account (see PRV).
- **FR-CUS-7 (SHOULD):** Let the customer mark a credential as lost and generate a new QR code.
- **FR-CUS-8 (MUST):** Show points that will expire soon, with the expiry date, as an in-app notice (reminder window proposed at 14 days).
- **FR-CUS-9 (MUST):** The customer app is available on the Apple App Store, Google Play and the web, with the same account on all three. Balances show as last synced when offline.
- **FR-CUS-10 (v1.1):** Show rewards a customer can afford, per store, and store offers and discounts.
- **FR-CUS-11 (v1.1):** Show alerts and offers in an inbox, with push notifications for opted-in customers.
- **FR-CUS-12 (v2):** Gift or transfer points, buy or send gift cards.

5.3 STO: store dashboard

Profile and staff

- **FR-STO-1 (MUST):** Owner edits store name, address, contact details, opening info, and logo.
- **FR-STO-2 (MUST):** Owner and manager invite staff by email, assign a role, and deactivate staff. Deactivation takes effect immediately, including on active sessions.
- **FR-STO-3 (MUST):** Every staff action that changes points is attributed to a staff member.

Points rules

- **FR-STO-4 (MUST):** Owner or manager sets points per euro, an optional minimum purchase, an optional maximum points per transaction, and the rounding rule (default: round down).
- **FR-STO-5 (MUST):** A change to the rule applies to future purchases only and records the date it took effect.
- **FR-STO-6 (SHOULD):** A points-rule preview lets the owner enter a sample amount and see the points result.
- **FR-STO-7 (MUST):** Points expire 3 months after the day they were earned, for every store, and the oldest points are used first. Stores cannot change this period in the MVP; a store-level setting is a later option.

Cashier screen

- **FR-STO-8 (MUST):** A dedicated, simple Cashier screen, installable as a web app on a phone or tablet, with a camera scanner, a numeric amount field, and a clear confirm button.
- **FR-STO-9 (MUST):** After scanning, show the customer's first name. Online, also show the current balance at this store; offline, never show a balance as if it were current.
- **FR-STO-10 (MUST):** Show a clear success, "Saved offline", or error state; never leave the cashier unsure whether an award was recorded.
- **FR-STO-11 (MUST):** Before the cashier confirms, show the amount and the points it will award. There is no undo after confirming.
- **FR-STO-12 (SHOULD):** Look up a customer by email when the QR code cannot be scanned (online only).

Offline use and devices

- **FR-STO-23 (MUST):** The Cashier screen opens and awards points without a connection once it has been loaded and synced at least once on that device.
- **FR-STO-24 (MUST):** Show the sync state at all times (online, offline, number of awards waiting) and the time of the last successful sync.
- **FR-STO-25 (MUST):** Owner or manager registers, names and revokes cashier devices. A revoked device is blocked at its next sync, and its unsent awards are held for review.
- **FR-STO-26 (MUST):** Show a list of rejected offline awards with the reason, and let the owner dismiss each one after review.
- **FR-STO-27 (SHOULD):** Show the last 10 awards made on this device, read-only, with their sync status.

Customers

- **FR-STO-13 (MUST):** Customer list with search, current balance, last visit, and total points earned and redeemed.
- **FR-STO-14 (MUST):** Customer detail page with ledger history and manual adjustment (see 4.5).
- **FR-STO-15 (MUST):** A store sees only customers linked to that store, and only the data needed for loyalty (name, contact if the customer consented, balance, history).

Reports

- **FR-STO-16 (SHOULD):** Basic report: customers enrolled, points issued and redeemed, awards per day, and top customers, for a selected date range.
- **FR-STO-17 (COULD):** Export report data as CSV.

Products, discounts and campaigns [v1.1]

- **FR-STO-18 (v1.1):** Add, edit and remove products; set a discount on any product with start and end dates.
- **FR-STO-19 (v1.1):** Create rewards with a point cost and optional limits.
- **FR-STO-20 (v1.1):** Create and schedule alerts to enrolled customers who opted in, with audience filters (for example, all customers or customers with a minimum balance).
- **FR-STO-21 (v1.1):** View alert results: sent, delivered, opened, and visits or redemptions afterwards where measurable.

Subscription

- **FR-STO-22 (MUST):** Owner views plan, next billing date, invoices, and payment status, and updates the payment method.

5.4 ADM: admin panel

- **FR-ADM-1 (MUST):** Create, edit, suspend, reactivate and offboard stores, with the status lifecycle in 4.1.
- **FR-ADM-2 (MUST):** Send, resend and revoke owner invitations.
- **FR-ADM-3 (MUST):** Search stores, staff and customers; open a record to see profile, status, links, and recent activity for support.
- **FR-ADM-4 (MUST):** Support actions (adjust points, apply a fraud clawback, deactivate a credential or device, merge duplicate customers) require a reason and are written to the audit log.
- **FR-ADM-5 (MUST):** View and filter the audit log across the platform.
- **FR-ADM-6 (MUST):** View each store's subscription and billing state, with the ability to extend a trial or pause billing.
- **FR-ADM-7 (SHOULD):** Platform dashboard: stores by status, new stores, total customers, awards per day, failed payments.
- **FR-ADM-8 (SHOULD):** A support notes field on each store and customer, visible only to admins.
- **FR-ADM-9 (SHOULD):** Admin tools respect the same privacy rules: customer detail is shown only as far as needed for support.

5.5 PTS: points engine

- **FR-PTS-1 (MUST):** Points = purchase amount in euros times the store's points-per-euro rate, rounded using the store's rounding rule (default: round down to a whole number).
- **FR-PTS-2 (MUST):** If the amount is below the store's minimum purchase, award zero points and tell the cashier why.
- **FR-PTS-3 (MUST):** If the result exceeds the store's maximum per transaction, award the maximum and show that it was capped.
- **FR-PTS-4 (MUST):** Each award request carries an idempotency key, so a retry, a double tap, or a resend after an offline period never awards twice.
- **FR-PTS-5 (MUST):** Write the ledger entry and update the balance in one atomic step; if either fails, neither is saved.
- **FR-PTS-6 (MUST):** Reject online awards for suspended or offboarded stores and for revoked credentials, with a clear message. Offline awards are judged at sync on the state at the scan time (FR-PTS-11).

- **FR-PTS-7 (SHOULD):** Warn the cashier when the same customer received an award at this store for the same amount within a short window (default: 2 minutes), and require confirmation. The check also works offline, on that device only.
- **FR-PTS-8 (MUST):** There are no reversals. A balance can never go below zero, except through an admin fraud clawback.
- **FR-PTS-9 (v1.1):** Redemption checks the balance, uses the oldest points first, writes a redeem entry, and rejects it if points are insufficient.
- **FR-PTS-10 (MUST):** Points expire 3 months after the day they were earned. Each earn entry records its expiry date and its remaining points, and a daily job writes an expiry entry for each batch that has run out of time.
- **FR-PTS-11 (MUST):** An offline award keeps the time of the scan. Its expiry date is counted from that time, and it is accepted at sync if the credential, the store status and the offline limits were valid at that time.
- **FR-PTS-12 (MUST):** A one-time QR proof can be used once per store. A second use is rejected, including when two devices send it at sync.

Worked example: the store rule is 1 point per euro, rounded down. The cashier enters 23.80 euros. The customer receives 23 points.

5.6 CRD: credentials

- **FR-CRD-1 (MUST):** Every customer has an app QR credential created at sign-up.
- **FR-CRD-2 (MUST):** The QR code can be checked without a connection. Proposed design: it carries a credential signed by the platform (customer, credential and the phone's public key, valid for about 7 days and refreshed automatically whenever the app is online) plus a one-time proof created by the phone, valid for about one minute. The cashier device checks both locally. The final design is set in the architecture work.
- **FR-CRD-3 (MUST):** The system resolves any credential type to the same customer ID, so awarding logic does not depend on how the customer was identified.
- **FR-CRD-4 (SHOULD):** A customer or admin can mark a credential lost or revoked. Online checks reject it at once; offline devices learn of it at their next sync, so the credential validity period limits the exposure.
- **FR-CRD-5 (v1.1):** Physical NFC card credential: stores only an opaque identifier, never a balance; cryptographically protected against cloning; linked to a customer by activation in the app or by an admin. Offline checking of cards needs card keys on the reader side and is designed in that phase.

- **FR-CRD-6 (v2):** Apple and Google Wallet pass credentials, with NFC tap only where the platform's approval is granted.

5.7 BIL: billing

- **FR-BIL-1 (MUST):** Each store has a monthly subscription with a plan, a start date, and an optional trial.
- **FR-BIL-2 (MUST):** Payments are processed by an external payment provider; BOKPOINTS stores no card numbers.
- **FR-BIL-3 (MUST):** Issue an invoice for each billing period that shows the correct Finnish VAT treatment.
- **FR-BIL-4 (MUST):** Failed payments follow this schedule, counted from the first failed payment: automatic retries, a reminder to the owner at day 7, the store Paused at day 14, and the store Suspended at the end of the first month (day 30). Paying at any point restores the store at once.
- **FR-BIL-5 (MUST):** A Paused store cannot create or change campaigns, rewards, products or points rules, but can still award points. A Suspended store cannot award points and its dashboard is read-only except for billing. Customers can always see their balances and history, and points still expire.
- **FR-BIL-6 (SHOULD):** Owner can change plan or cancel; cancellation takes effect at the end of the paid period.
- **FR-BIL-7 (COULD):** Charge separately for NFC cards.

5.8 NTF: notifications

- **FR-NTF-1 (MUST):** In the MVP, transactional messages go by email only: verification codes, invitations, security notices and billing reminders. SMS is not used.
- **FR-NTF-2 (MUST):** Customer notices (a new award, points expiring soon) appear in an in-app notice area.
- **FR-NTF-3 (v1.1):** Push notifications and store alerts go only to customers who opted in for that store, and every message gives an easy way to opt out.
- **FR-NTF-4 (v1.1):** Customers can control channels (push, email) and mute a single store.
- **FR-NTF-5 (v1.1):** Limit how many marketing messages a customer can receive per store per period (limit to be decided).
- **FR-NTF-6 (v2):** SMS as an extra channel once a cost plan exists. SMS fees are normally billed to the sender, so BOKPOINTS would carry them unless they are passed on to stores.

5.9 API: API and integrations

- **FR-API-1 (MUST):** The Cashier screen uses the same internal API as any future integration: identify customer, award points (including batches of offline awards), and get balance.
- **FR-API-2 (MUST):** Every API call is authenticated, scoped to one store, and permission-checked.
- **FR-API-3 (MUST):** The API is versioned and documented from the start.
- **FR-API-4 (v2):** Store-level API keys and webhooks for POS integrations, with rate limits and per-key revocation.
- **FR-API-5 (v2):** Webhooks notify a store system when points are awarded or reversed.

5.10 AUD: audit and fraud

- **FR-AUD-1 (MUST):** Record in the audit log: store creation and status changes, staff changes, rule changes, manual adjustments, fraud clawbacks, credential and device changes, rejected offline awards, and admin support actions.
- **FR-AUD-2 (MUST):** Audit entries cannot be edited or deleted through the product.
- **FR-AUD-3 (SHOULD):** Flag unusual patterns for review: many awards by one cashier in a short time, repeated large amounts, many adjustments, or repeated failed scans.
- **FR-AUD-4 (SHOULD):** Rate-limit scanning and awarding per device and per cashier.
- **FR-AUD-5 (SHOULD):** Limit the exposure of offline use: cap the amount per offline award and the number of awards waiting per device, and flag devices whose clock differs from server time.

5.11 PRV: privacy and consent

- **FR-PRV-1 (MUST):** Show the terms and privacy notice at customer sign-up and record the version accepted and the time.
- **FR-PRV-2 (MUST):** Marketing consent is separate from the terms, off by default, recorded per store, and can be withdrawn at any time.
- **FR-PRV-3 (MUST):** A customer can export their data and delete their account. On deletion, personal data is removed and ledger entries are kept in anonymized form for store accounting.
- **FR-PRV-4 (MUST):** A store can see only its own customers' data and cannot see other stores' balances or history.

- **FR-PRV-5 (MUST):** Delete unclaimed pending-customer records, including their email addresses, after 1 year. Define other retention periods for inactive accounts and apply them automatically.
- **FR-PRV-6 (MUST):** Log access by admins to customer data.

5.12 LOC: localization

- **FR-LOC-1 (MUST):** Finnish and English (proposed) in all customer, store and admin screens and in messages.
- **FR-LOC-2 (MUST):** Currency in euros, with Finnish number and date formats.
- **FR-LOC-3 (SHOULD):** Text is stored in translation files so more languages can be added later without code changes.

6. Screen inventory (MVP)

Customer app

- Sign up and sign in
- Home with QR code
- Stores, balances and expiring points
- Store detail and history
- Profile and consent settings
- Data export and account deletion

Store dashboard (web)

- Sign in and setup wizard
- Cashier screen (scan, amount, confirm, result, offline and sync status)
- Customer list and customer detail
- Points rules
- Staff, cashier devices and rejected offline awards
- Store profile
- Basic reports
- Subscription and invoices

Admin panel (web)

- Sign in

- Stores list and store detail
- New store and invitation
- Customer and staff lookup
- Audit log
- Billing overview
- Platform dashboard

7. Edge cases and error states

Situation	Required behavior
QR code expired (proof older than about a minute, or credential past its validity)	Cashier sees "Code expired, ask customer to refresh"; no points awarded
QR code replayed, copied from a screenshot, or credential revoked	Rejected; online at once, offline at sync; event logged
Network lost during award	Award is saved offline and sent later; the same idempotency key prevents a double award
Cashier enters the wrong amount	The confirm step shows the points first; after confirming there is no undo. Owner or manager can adjust points, never below zero
Customer scanned at a suspended store (online)	Rejected; customer unaffected
Store becomes suspended while a device is offline	Awards scanned before the suspension are accepted at sync; later ones are rejected and shown to the owner
Same one-time QR proof used on two offline devices	First to sync is accepted; the second is rejected and reported
Offline award rejected at sync	Customer gets no points; owner sees it in the rejected-awards list and can add points manually if the purchase was genuine
Same customer awarded twice for the same amount in 2 minutes	Warning and confirm step

Situation	Required behavior
Two cashiers award the same customer at once	Both succeed in order; ledger remains consistent
Customer has no account yet	Cashier can register by email (see 4.4)
Customer deletes their account	Personal data removed; stores see the customer as removed; anonymized ledger kept
Store rule changed after a purchase	Past awards are unchanged; an offline award uses the rule version cached on the device, and the ledger records which version applied
Removal or clawback would take a balance below zero	Blocked, except for an admin fraud clawback
Camera permission denied	Cashier is prompted to allow it, or uses lookup by email (online only)
Cashier device lost or stolen	Owner revokes it; unsent awards from it are held for review
Customer phone clock is wrong	The proof check fails; customer is asked to correct the time
Duplicate customer accounts	Admin merge tool with audit entry

8. Non-functional requirements

- **FR-NFR-1 (MUST) Speed:** a scan-to-confirmation round trip should feel immediate. Proposed target: under 3 seconds on a normal mobile connection, measured at the 95th percentile. An offline confirmation appears on the device within about one second (proposed).
- **FR-NFR-2 (MUST) Reliability:** the award flow is the most critical path; plan monitoring and alerts for it from the first release. The offline queue survives app restarts and device reboots, and retries with backoff until the server confirms each item.
- **FR-NFR-3 (MUST) Security:** encrypted traffic and stored data; hashed secrets; protection against common web attacks; least-privilege access; secrets kept out of code.

- **FR-NFR-4 (MUST) Data integrity:** automatic backups with a tested restore, and a ledger that can always be reconciled to balances.
- **FR-NFR-5 (MUST) Devices:** customer apps for current iOS and Android versions from the App Store and Google Play, and for current Safari and Chrome on the web; store and admin dashboards in current desktop and mobile browsers, with the Cashier screen installable on phones and tablets.
- **FR-NFR-6 (SHOULD) Accessibility:** follow WCAG 2.1 AA practices; readable contrast and large tap targets on the cashier screen.
- **FR-NFR-7 (MUST) Observability:** structured logs, error tracking, and a way to trace one award end to end for support, including its offline queue and sync history.
- **FR-NFR-8 (SHOULD) Privacy by design:** collect only the data needed; no customer data in logs.
- **FR-NFR-9 (MUST) Testing:** automated tests for the points engine, expiry, permissions, the award flow and offline sync; a manual test checklist for each release.
- **FR-NFR-10 (MUST) Offline security:** data stored on a cashier device is encrypted and limited to what the till needs; no customer list is kept on the device.

9. MVP acceptance criteria

AC-1 Award points

- Given an active store with a rule of 1 point per euro (round down) and a customer with a valid QR code,
- when the cashier scans the code and enters 23.80,
- then 23 points are added once, the customer's balance increases by 23, and both screens show the confirmation.

AC-2 No double award

- Given an award request was sent and the response was lost,
- when the same request is retried with the same idempotency key,
- then the customer's balance increases only once.

AC-3 Store isolation

- Given a customer has points at Store A and Store B,
- when a Store A manager opens the customer,
- then only Store A's balance and history are visible.

AC-4 Permissions

- Given a cashier is signed in,
- when they try to open the points rules or make a manual adjustment,
- then the server refuses and the attempt is logged.

AC-5 Suspended store

- Given a store is suspended for non-payment,
- when a cashier tries to award points,
- then the award is refused with a clear message, and customers can still see their existing balance.

AC-6 Expiry

- Given a customer earned 23 points at a store on 2 June,
- when the date reaches 2 September,
- then an expiry entry of minus 23 points is written and the balance drops by 23.

AC-7 Customer deletion

- Given a customer requests account deletion,
- when the deletion completes,
- then their personal data is removed, their credentials stop working, and ledger entries remain without personal identifiers.

AC-8 Admin onboarding

- Given an admin creates a store and sends an invitation,
- when the owner accepts it and completes setup and billing,
- then the store becomes Active and its cashiers can award points.

AC-9 Award offline

- Given a cashier device that synced earlier and now has no connection, and a customer with a valid QR code,
- when the cashier scans the code and confirms 23.80, and the connection later returns,
- then the award shows "Saved offline", is sent at sync, is written to the ledger once with its original scan time, and the customer's balance rises by 23.

AC-10 Offline award rejected

- Given an offline award made after the store was suspended, or with a QR proof that was already used,
- when the device syncs,
- then the award is rejected, no points are written, and it appears in the store's rejected-awards list with the reason.

AC-11 No negative balance

- Given a customer has 5 points at a store,
- when a manager tries to remove 8 points,
- then the removal is refused; and when an admin applies a fraud clawback of 8 points with a case note, the balance becomes minus 3 and the link is marked under fraud review.

10. Later phases (outline)

v1.1

- Rewards catalog and redemption at the till (oldest points first).
- Products and discounts in the store shop.
- Store alerts and campaigns with consent, limits and results.
- Push notifications for opted-in customers.
- Physical NFC loyalty card and its activation flow, including offline checking of cards.

v2

- Gifting and transferring points, with limits and fraud controls.
- Gift cards (needs a regulatory review first).
- Restock, last-stock and seasonal sale alerts.
- Apple and Google Wallet passes, with NFC tap where approved.
- Store API keys, webhooks and the first POS integrations.
- SMS as an extra notification channel, with a plan for its cost.
- Store-level expiry settings.
- Possible universal points (needs settlement design and legal review).

11. Decisions made and open points

Decision	Choice
Points expiry	Points expire 3 months after the day they were earned, in the MVP (moved up from v1.1)
Reversals and negative balances	No reversals. Balances never go below zero, except an admin-only fraud clawback. Manual adjustments stay, never below zero
Pending customers	Included in the MVP; registered by email; unclaimed records deleted after 1 year
Customer app	Apple App Store, Google Play and web
Failed payments	Reminder at day 7, store Paused at day 14, Suspended at day 30
Offline awarding	Required in the MVP: QR checked on the device, awards queued and synced later
Messages and cost	Email verification and in-app notices in the MVP; SMS later
Pilot stores' needs	Scope frozen; pilot requests go to a backlog reviewed after the pilot

Open points

1. **Expiry meaning (confirmed):** this FRD reads "every 3 months" as each batch of points expiring 3 months after it was earned.
2. **"Pause" (confirmed):** this FRD defines Paused as no changes to campaigns, rewards, products or rules, while awarding continues.
3. **Manual removal (confirmed):** owners and managers can still remove points by adjustment (never below zero).
4. **Offline limits:** the cap per offline award, the queue size and the 7-day credential validity are proposals, to be set in the architecture work.
5. **Fraud clawback process:** who decides a case, and what the case note must contain.
6. **Pending-customer data:** keeping an email address for a year without an account needs a privacy review of the lawful basis.
7. **Expiry reminder:** the 14-day window is a proposal.

8. **SMS costs:** SMS fees are normally billed to the sender, not the recipient. A later plan must decide whether BOKPOINTS absorbs them or passes them to stores.
9. **App store accounts:** developer accounts and review lead time for the iOS and Android apps are launch tasks.

12. Traceability to the PRD

PRD goal	FRD modules
Fast points at the till	STO (Cashier screen), PTS, CRD, API
One view of points across stores	CUS
Tools for stores	STO, PTS, NTF (v1.1)
Full admin control	ADM, AUD, BIL
Extendable to NFC, wallets and POS	CRD, API
Finland and EU readiness	PRV, LOC, BIL

13. Next steps

1. Review these changes and confirm the open points in section 11.
2. Align the PRD with these decisions.
3. Decide the technology stack (cross-platform apps and an offline-capable web app), then write AGENT.md.
4. Start the handoff document and the decision log.